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Practices in motion: expansive learning and the flow of digital collaboration in transforming healthcare

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ABSTRACT

This paper explores how healthcare professionals learn through the introduction of a digital wound care application, focusing on how new forms of collaboration and practice emerge in response to digitalisation. Situated within the context of a formative intervention study using the Change Laboratory method, the research draws on qualitative data from seven sessions involving interprofessional participants. The aim was to investigate how learning unfolds when established routines are disrupted, and how professionals engage in expansive learning across organisational boundaries. Rather than simply adopting a new tool, participants reconfigured their object of care and reorganised their activity system through processes of reflection, negotiation, and experimentation. Drawing on expansive learning theory and complemented by flow theory, the study examines how learning is enacted through the temporal (timing), perceptual (attentionality), and experiential (undergoing) dimensions of practice. The findings highlight that learning is not a linear process of adaptation, but an emergent, situated transformation driven by contradictions within existing routines. The study contributes to understanding professional learning as a dynamic and practice-based process, where technological innovation, organisational change, and knowledge development are co-constituted.

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Introduction

As healthcare systems undergo rapid digital transformation, the implementation of new technologies increasingly intersects with evolving professional roles, inter-organisational collaboration, and shifts in established work practices (e. g. Altman Ferreira 2025; Bergey, Goldsack, and Robinson 2019; Gjellebæk et al. 2020; McNulty and Ferlie 2002). While much research has focused on the technical and organisational challenges of digitalisation, less attention has been paid to how professionals actually learn in and through these implementation processes, particularly when learning extends beyond existing routines and institutional boundaries (Ghosh et al. 2023; Wessel et al. 2021).

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In this context, the theory of expansive learning, as developed by Engeström (1987, 2001), offers a productive lens for understanding how new forms of knowledge and practice emerge in response to systemic contradictions. Expansive learning occurs when individuals and groups collectively question existing practices, envision new models, and gradually construct and internalise novel patterns of activity. It is especially relevant in complex, multi-actor environments – such as healthcare – where transformation often requires boundary crossing between organisations, professions, and knowledge domains (Engeström and Pyörälä 2021). The implementation of digital applications in healthcare settings can trigger such boundary-crossing processes, especially when they challenge established divisions of labour or require joint problem-solving across care levels. How a digital application can facilitate knowledge sharing between different healthcare professionals has been shown in e. g. Aggestam and Svensson (2025). Svensson et al. (2023) also show that use of digital applications changes work processes in healthcare for the professionals, where they learn new ways of working.

Despite growing interest in digitalisation in healthcare, there is limited understanding of how professionals engage in and make use of emerging opportunities for learning in practice. Research often emphasises either individual skills or system-level impacts, overlooking how professionals negotiate collaboration when digital tools disrupt routines. This paper addresses that gap by examining how learning unfolds during the introduction of a digital application across organisational boundaries. By drawing on expansive learning and flow theory, this paper highlights how learning is mediated through the interplay of a digital application and interprofessional collaboration, enabling the development of new, situated ways of knowing and acting in practice.

Theoretical framework

Work is a domain of activity, where driving forces are prevalent as a system. Engeström (1999) has developed the third generation of activity theory. Minimal for the third generation of activity theory is two interacting activity systems. This implies boundary crossing between different activity systems. Different actors meet and interact to go beyond each evident limit. The activity theory provides conceptual tools to understand dialogue, multiple perspectives, and networks of interacting activity systems.

Activity theory and its five principles

The activity system is object-oriented, collective and artefact-mediated, and can be seen in relations to other activity systems. This is taken as the unit of analysis. Activity systems generate actions and operations, that are both individual and group goal-directed actions (Engeström 2001).

An activity system is a community of multiple points of view, interests and traditions. There is also a certain division of labour that creates different positions for the actors. The actors carry their own histories, that are diverse in their character. The multi-voicedness is multiplied when activity systems are interacting with each other. It is also a source of

troubles, contradictions, and innovations, that demand action of translation and negotiation (Engeström 2001). Expansive learning is an 'inherently multi-voiced process of debate, negotiation and orchestration' (Engeström 2014, 5).

Problems and potentials in activity systems need to be understood against their own history, therefore it is important to recognise the historicity in activity systems. Therefore, local history of the activity and its objects, in relation to ideas and tools that have shaped the activity, needs to be studied. Any local organisation has concepts, procedures and tools employed and accumulated in the local activity in its history (Engeström 2001).

Contradictions in an organisation are sources of change and development. However, contradictions are not the same as problems or conflicts. Contradictions are tensions within and between activity systems, that have been historically accumulated. New elements that are added into an activity system, and old elements can collide with the new ones. This generates disturbances and conflicts, as well as attempts to changes and innovations (Engeström 2001).

There are movements in cycles of qualitative transformations in activity systems, called expansive cycles. When contradictions are aggravating, individuals begin to question and deviate from the established norms. Sometimes, this escalates into collective change efforts. The object and the motive of the activity are then reconceptualised in comparison to the previous mode of the activity, and an expansive transformation is accomplished (Engeström 2001).

Expansive learning

Traditionally, learning has been seen as the acquisition of stable knowledge or skills by individuals. However, such stability is often lacking in practice. As Engeström (2001) notes, people '*are all the time learning something that is not stable, not even defined or understood ahead of time*' (137). Transformations are ongoing in both personal and organisational contexts, and learning involves adapting to new, not-yet-existent forms of work (Baygi, Introna, and Hultin 2021). Expansive learning refers to this process, where learners collectively reshape their activity system and create new cultural patterns of work (Engeström 2001; Engeström and Sannino 2010).

Rather than focusing on well-bounded communities of practice, expansive learning takes place in heterogeneous networks, involving diverse institutions, actors, and technologies. These configurations trigger contradictions and tensions that can fuel learning (Engeström, Engeström, and Vähäaho 1999). Motivation arises not from formal instruction but from participation in collaborative practices that are expected to yield value (Lave and Wenger 1991).

A double bind emerges when persistent contradictions cannot be solved within existing structures. These moments, though challenging, can provoke reflection and collective inquiry. As Engeström (1987) describes, a double bind is '*a social, societally essential dilemma which cannot be resolved through separate individual actions alone – but in which joint co-operative actions can push a historically new form of activity into emergence*' (165). Learning in such contexts is not about acquiring fixed knowledge, but about transforming the object, purpose, and organisation of activity through cycles of modelling, experimentation, and reconfiguration (Engeström 2001).

Flow of actions

Flow theory highlights how actions and interactions unfold as continuous, dynamic processes shaped by timing, attention, and ongoing adaptation (Baygi, Introna, and Hultin 2021).

Timing refers to the temporal alignment of actions within a shared practice, focusing on the 'right' or opportune moments (kairos) rather than linear clock time (chronos). It captures how different activities, each with their own rhythm and pace, intersect in ways that create openings for new possibilities. In social and practical contexts, this means that actions do not follow a single, fixed schedule but flow according to the needs and dynamics of the situation (Orlikowski and Yates 2002). When these flows align appropriately, transformation becomes possible.

Attentionality is the skilful, often implicit, orientation of focus towards possibilities for action as they arise in ongoing practice. It is more than conscious awareness or deliberate planning; it involves a sensitivity to subtle cues, rhythms, and changes within a situation. For example, healthcare professionals develop a heightened perception of clinical signs and shifts in patient conditions, responding intuitively to emergent demands (Baygi, Introna, and Hultin 2021). This attentional focus is fluid and relational, constantly adjusting to multiple simultaneous demands and flows.

Undergoing captures the experiential and affective dimension of engaging with ongoing activity flows. It refers to how individuals actively participate in and are transformed by these flows through a dynamic interplay of leading and following, described as doing-in-undergoing. Rather than fixed individual agency, undergoing reflects how agency emerges through creative alignment and responsiveness to others and the environment – similar to how a surfer must attune to waves to move skilfully (Baygi, Introna, and Hultin 2021). The actualisation of action possibilities depends on this openness to being influenced and changing in the moment. Together, timing, attentionality, and undergoing reveal how learning and practice evolve as fluid, embodied, and relational processes embedded in sociotechnological systems.

The challenge in learning wound healing

Healthcare in Sweden is organised across two regional levels – primary care and hospitals – and one municipal level. Patients with wounds usually seek care at their local primary healthcare centre, where nurses primarily manage wound assessment and treatment, often with support from physicians. When wounds are especially difficult to heal, referrals to hospital specialists are made to provide more advanced care.

In municipal home healthcare, nurses visit older adults or individuals with disabilities to provide wound care at home. These municipal nurses can consult physicians at primary care centres if complications arise. Should the wound worsen, patients may be referred to hospitals for specialised treatment.

Effective wound healing has great potential to improve patients' quality of life; however, many wounds remain chronically unhealed. Wound healing is complex and influenced by multiple interacting factors. It is not a matter of luck but

requires specific knowledge. Although knowledge about wound healing exists, healthcare professionals in municipalities and primary care must master a wide range of topics, resulting in fragmented expertise regarding different types of wounds. This fragmentation was a key motivation for introducing the digital application.

Wound care often involves several professionals over time, complicating the monitoring of the healing process. Even when information is recorded in medical records, understanding the wound's status and treatment history can be challenging.

Research methods

This paper is based on a qualitative action case study, performed as a formative intervention study as a Change Laboratory. This is a method used to involve practitioners in research together with researchers (Sannino and Engeström 2017). In the study a digital application for health-care professionals concerning wound support was used. The formative intervention was aimed at generating and implementing new working methods using a mobile application that provides support with wound assessment and choosing the right treatment for a specific wound, to improve efficiency in patient assessment and treatment. The mobile application consists of two parts, a knowledge library with information about wounds, and a structured wound assessment and wound treatments tool, including recommendations of bandages for dressing.

The intervention included seven regular digital Change Laboratory sessions (Engeström, 2018), held for one hour every second week during a period. The health-care professionals were provided with a digital application for wound assessment and treatment. In between the sessions, the health-care professionals were asked to use the new digital application in practice.

In total 10 health-care professionals, together with four people from staff at the company developing the digital wound support application and three researchers participated in the study. Two of the health-care professionals work at the primary health care centre organisation and eight work at the municipality health-care organisation. This enabled collaborative development beyond authoritative organisational constraints (Sannino and Engeström 2017). Thus, the health-care professionals belonged to two different communities of practice. From the company one project manager, one business manager, one IT director and one global marketing manager participated. One of the researchers led each meeting.

At each of the sessions one of the participants was asked to describe an example from work practice, where a hard-to-heal wound was challenging and where the digital application could be used to facilitate the work. The problematic situation from the work practice, where the health-care professional was confronted with wounds hard to heal, composed the first stimulus, as this is the problem to be tackled. The digital application became the second stimulus, as the mediating artefact, in the double stimulation in this Change Laboratory (Sannino, Engeström, and Lemos 2018). The digital application, as the second stimuli, was aimed to support the health-care professional to gain control of the problematic situation in practice. Thus, the digital application became a mediating and agentive artefact in work practice, and support in learning.

The collaborative sessions were recorded, and notes were taken. During the thematic analysis the recorded sessions were listened to, and excerpts were chosen to illuminate the learning (Braun and Clarke 2006).

Results

Who and where are the professionals of learning?

In this case nurses from a primary healthcare centre and from a municipality, together with representatives from a company, participated in six workshop sessions in the boundary crossing laboratory, conducted online due to pandemic restrictions. The three interconnected activity systems are shown in Figure 1.

The individual practitioners within this healthcare context have obtained an exam as a registered nurse. There are also some of the nurses who have further been trained as specialist nurses. During the education the nurses have adopted knowledge and skills due to different injuries and illnesses, as well as wound assessment and treatment. The nurses work at either the primary healthcare centre or at the municipal healthcare unit. The nurses, to a different extent, meet patients with wounds that are hard to heel, and then they also adopt skills and experiences due to different types of wounds. However, it is quite challenging to work with wounds, as many of the wounds are hard to heel. The nurses also often work alone, and then they have to assess and treat wounds based on their own knowledge, skills and experiences.

The representatives from the company work within the context of a digital application that should support assessment and treatment of wounds. The company both develop the digital application as well as they are providing dressings for wounds to the market.

The learning by the professionals in the boundary crossing laboratory was based on and triggered by situations with wounds where healthcare professionals had faced problems and where the digital application could be a support. There was no predefined agenda for learning, but the learning was based on each professional's experiences in practice in working with wounds and with the digital application in their work with wounds. There were no top-down guidelines or structures from the management, nor from the researchers, on what content should be discussed.

The three activity systems were implemented into the boundary crossing laboratory. The voices from the different participants in the ongoing discussions indicated the multi-voicedness of the interaction in the sessions. The healthcare professionals' voices confirmed the difficulties in discussing and sharing knowledge in their everyday practice and opened up for using the application for knowledge sharing and learning as working with healing of wounds is a difficult task. The digital application can support expansive learning, as new patterns of activity can be identified, when individuals as learners are engaged in using the application and using the results from previous assessments that are registered in the application. Excerpt 1 is an example of this.

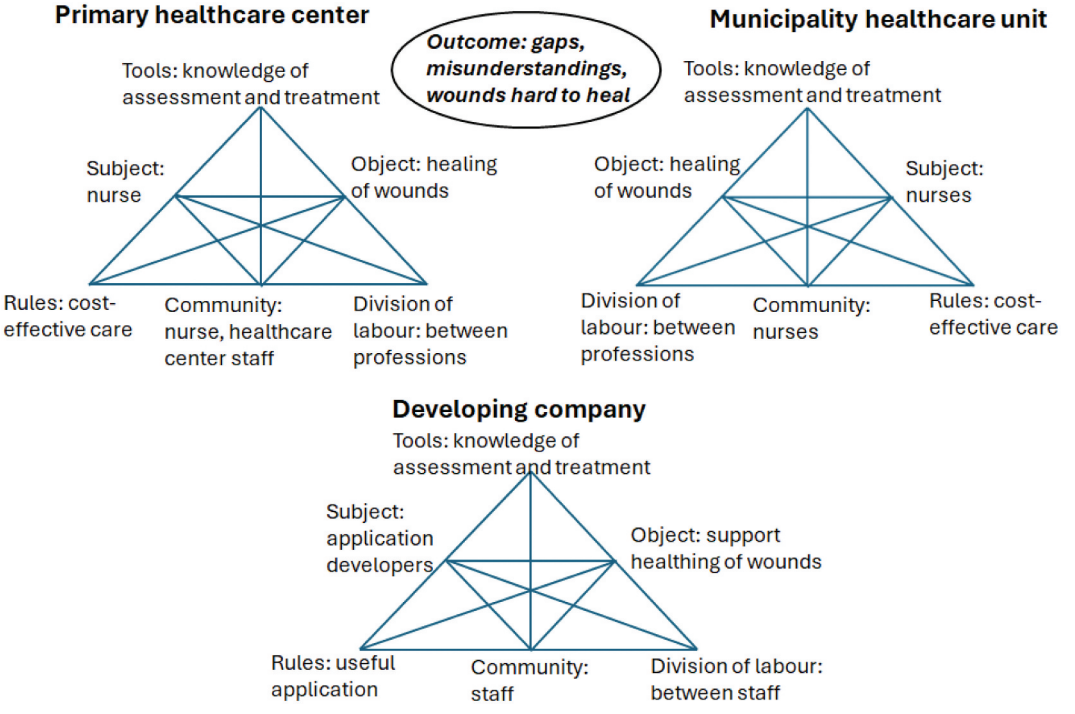


Figure 1. The three interconnected activity systems in the case.

Excerpt 1 (Boundary Crossing Laboratory, session 2).

Speaker 5 (municipality nurse)	Should we have the tool to discuss with each other, do you think? That we, that we compare wounds? If we have, like, as someone said there, if we're not that used to dressing wounds, that we can then, by having taken, taken the photo there and can discuss more easily. Is that something you have?
Speaker 1 (nurse primary hcalth center)	Yes, but then, you're thinking that you can do that through the app, right?
Speaker 5	No, but I think that if you have, if you have it like that, so that you have, since the app you don't talk to each other and it shuts down, if you say so, afterwards. But that with the guidance you get, you can have a dialogue . . .. Maybe someone shares a picture with someone. –This is what it looks like, what do you think?
Speaker 1	What do you others think about that?
Speaker 7 (municipality nurse)	In the past, people who were more knowledgeable about wounds, who were educated, etc., were asked. "What should I do with this wound?" But of course, if you had had one, you could use the app and show the picture.

Individuals from the different practices discussed wound healing from their own perspectives and knowledge. From different perspectives different aspects are important. For healing of wounds, it is important to take a more holistic perspective, however, the representative from the company developing the application is more concentrating on assessment of the wound. Therefore, contradictions in the three interconnected activity systems are made clear in the boundary crossing laboratory, as shown in Excerpt 2.

Excerpt 2 (Boundary Crossing Laboratory, session 2).

Speaker 3 (municipality nurse)	We haven't had time, but there's so much else around a wound, you think. Just with diet and with swelling and. So there's so many underlying diseases they have, so you have to get rid of a lot of fluid and so on. So it's not just a wound, there's so much around each person too that's important to talk about.
Speaker 6 (company staff)	Mm.
Speaker 3	You can't just look at the wound, eh, and then solve it, I feel. Without that, so much else too. Or am I thinking wrong? Or am I on the wrong track?
Speaker 1 (nurse primary hcalth center)	Yes, I think exactly like you Betty (speaker 3) and I have actually written it as a small point here on what one could include in the knowledge library. It is precisely this as well as this. What should I say? General factors that can affect healing? I mean nutritional status, diseases, medications, sleep, pain, stress. That this holistic approach specifically around wounds and, or person-centered care. It is also, one could have as a section in the knowledge library I think, there are so many.
Speaker 6	Mm. I'm looking around a bit here now in the knowledge library. Because then we actually have, that's right, because we have, we're looking into things and how to assess a certain type of wound and what to do. But then we choose not to include, for example, the nutrition part, as we know, for example in connection with pressure ulcers.

Table 1 show the multi-voicedness in the related activity systems.

Table 1. Activity systems and their multi-voicedness.

	Activity system as unit of analysis	Multi-voicedness
Who are learning?	Interconnected activity systems Primary care center, municipality care, application developing company	Voices of primary care, municipality care, application developing company

Why do the professionals learn?

Traditionally, the healing of wounds has been stable practice. Healthcare professionals have got basic knowledge about wound healing during their studies and are then gaining competence during their professional practice. However, the practice is facing problems when wounds sometimes are hard to heal, or in some cases, when wounds seem almost impossible to heal. Of course, this means great suffering for the patients, as wounds most often are very painful. Just because wound healing is that challenging for the healthcare professionals, the first initiative was taken from a nurse (speaker 1) from the primary healthcare centre, to try the digital application, and to collaborate for knowledge sharing in this boundary crossing laboratory. The aim of the introduction of the application is to structure and support the work, and to improve assessment and treatment in wound healing. As the first initiative was taken from speaker 1, a nurse from the primary healthcare centre, she took a leading subject position in the boundary crossing laboratory.

The wound healing practice includes ambiguities, with contradictory views of different individuals. It is, for example, difficult to assess different aspects that have an impact on the wound, and different professionals have more or less knowledge and experience. The implementation of the digital application compasses a major transformation of the performance of the work practice. The digital application to a certain extent prescribing a procedure to assess a wound. However, what is included in the application can be interpreted in different ways. This was shown from the first session, in Excerpt 3.

Excerpt 3 (Boundary Crossing Laboratory, session 1).

Speaker 12 (nurse primary hcalth center)	Well, I assessed it after I had washed it then. But I think it's very difficult to put a percentage like that. I think I put 10% fibrin" and 90% red granulation there. It's a bit hard to say, I think.
Speaker 1 (nurse primary hcalth center)	Yes, I thought afterwards that I should have done as you did and made the assessment afterwards. Is there anyone else who. . . I wonder, you others who have made wound assessments, what have you thought about this?
Speaker 2 (municipality nurse)	I think you do it after washing so you can see what's left after you've washed it. What you have as a starting point.
Speaker 1	Yes.
Speaker 2	Or how do you say? Mary? (speaker 9)?
Speaker 9 (company staff)	That's the idea, yes.
Speaker 1	Exactly, yes, then I think it's sometimes difficult to see what is granulation and what is epithelialization, but maybe that's just me. . . yes, I think it's difficult to know the difference. . . I haven't googled it, but yes. . . How do you others feel?
Speaker 13 (municipality nurse)	I think that I am not that used to working with wounds because I work mostly in dementia care during the day and then I am out in home care in the evenings and weekends and there are no planned wound dressings in that way. So we have different skills in wound treatment.

In the work with wound healing there are also double binds. The dilemmas in wound healing cannot be solved with separate individual professionals. It is a process to heal wounds, and different professionals meet the patient during this flow of time for the healing process. It means that the way of working can be different from each individual professional, and that they do their own assessment, based on their own knowledge and competence. In using the application this double bind dilemma can be solved, as the

professionals need to use cooperative actions that form a new type of activity in their work of wound healing. The application can be used by the different professionals, with its prescribed process, with the structured documentation from previous meetings with the patient where the professional can clearly see the progress of the wound healing from time to time during the healing process, and with the application library that is accessible from the application. In this way the continuity, that is also critical for wound healing, can be improved, as exemplified in Excerpt 4.

Excerpt 4 (Boundary Crossing Laboratory, session 5).

Speaker 1 (nurse primary hcalth center)	So we've used Xylocaine gel before. Because we need to scrape some, and it's been good when we've done that. Eh, let's see here. I thought we'd just see. There! 2.9 × 0.9 then, was the length and width. Then, now, took, hit, I saw or I redressed this wound a week later I think it was. On, on Monday. And then it looks like this and then you can see here that it's kind of gotten a little wider here and maybe a little shallower, but it's kind of expanded, the wound. Eh, and you can also see that, you can also see that here that it's 3, all of a sudden 3.5 and 0.9 so it's gotten longer, the wound. Eh, and even the yellow fibrin part has gotten a little more. Eh, if you look there. Yes, it was, it was maybe about the same. Yes, now I don't know how this ended up? No, yes exactly, it's 40%. Yes, I had said 35. That. No exactly, 65. Yes, this was it. So it has increased. I think this was a little interesting because I, I'm not sure that if you didn't have. Now I'll see if I can find the wound again. I'm not sure that, if you didn't have this app so I'm not sure that you would have noted in the same way the, the change that happened.
Speaker 5 (municipality nurse)	Mm.
Speaker 1	So, it's not so easy to have a memory of what it looked like. Now I see that this has gotten bigger and then you start to reflect: –Why is it getting bigger? And then you kind of take a new approach to the wound. And I think, my feeling is anyway that, that you do it at an earlier stage, when you see so clearly that, I mean, the part for fibrin is increasing and that the size of the wound is increasing. Then, then you kind of see that yes, but we have to do something about this. Eh, we can look at the other wound then too, which is on the outer malleolus, looks like this. There you see that, yes, yellow necrosis 30 approximately and 70% I had estimated at that time to be red granulation. And it was 2.3 in length and 1.3 in width. And then then, next week it looks like this. Then I think there's a little more granulation, what's it called? Fibrin. And then I also assessed that then all of a sudden it's 55%. From 30 I think it was, right?
Speaker 5	Mm.
Speaker 1	Then it has also increased in length and width. So here too the wound has worsened. Yes, now I made a change. I, I tried, I had Intrasite gel here, but tried Iodosorb instead. We'll see, and then I had Mepilex dressing over then.
Speaker 12 (municipality nurse)	What was it before then?
Speaker 1	There was Mepilex and then there was Intrasite Gel.
Speaker 12 (municipality nurse)	Yes, okey.

In this case there exist internal contradictory demands. There is an internal demand within each healthcare organisation that they should be cost-effective, i.e. that they should use the cheapest materials from their point of view. However, wrong decision can be taken because cheaper material can cause delays in wound healing. Different materials, in combination could be needed, which means that wound healing will be a lot more expensive in the longer term, and maybe for another healthcare organisation. Therefore, as seen as a single assessment of a wound, a certain solution can be cheaper, but in the longer term it can be much more expensive. If more expensive material is used for a short time, it is still possible to reduce overall costs, as the wound can be healed more quickly. Then also the time used for assessments and treatments for the very same wound can be reduced, as is described in Excerpt 5.

Excerpt 5 (boundary crossing laboratory, session 4).

Speaker 1 (nurse primary hcalth center)	Because, because it was a little interesting if you want to hear a little bit. But we have it, we went through what kind of dressing should we have? And then you look at prices and everything like this. And then we chose. Allevyn, because it's a little cheaper. But what we noticed later.
Speaker 9 (company staff)	Mm.
Speaker 5 (municipality nurse)	Mm.
Speaker 1	But, what we noticed then is that we, we double-. Because we noticed this, that it's not soaking up, we put both Aquacel and Allevyn.
Speaker 5	Mm.
Speaker 1	So we double-, actually double-bonded.
Speaker 5	Mm.
Speaker 1	And then it won't be cheaper.
Speaker 9	No.
Speaker 1	So we had to, had to rethink that. No, no, even those who didn't dress wounds that often came to us: -You know, this dressing, it's not, it's like, it doesn't absorb properly. So then we thought: -No, but now we have to change it. So that, it's really like that, yes. You have to send it again and you try to look at it cost-effectively. But, but it's also like, how. Yes, it doesn't get cheaper if you have to use two, so to speak.
Speaker 5	No, and if you then need to change the dressing much more often, uh, it sticks and that, then the 2 kronor you have saved, uh, is quickly gone in a few seconds of extra dressing change, really.
Speaker 1	Yes, my goodness, and just not having to change a bandage once a week makes a huge difference.
Speaker 5	Yes, yes. A lot.
Speaker 1	For both patients and staff.

The historicity and contradictions in the interconnected activity systems are shown in Table 2.

What are the professionals learning?

By using the application's built-in camera to take wound images, healthcare professionals were able to engage in detailed discussions in the collaborative sessions about the condition of the wounds, based on specific visual features. This helped

Table 2. Historicity and contradictions in the interconnected activity systems.

	Activity system as unit of analysis	Multi-voicedness	Historicity	Contradictions
Who are learning?	Interconnected activity systems Primary care center, municipality care, application developing company	Voices of primary care, municipality care, application developing company		
Why do they learn?			Health professionals have basic knowledge about wound healing. Professionals are facing problems in practice when wounds sometimes are hard to heal.	Contradictions between using the cheapest material for treatment in a short term and the most effective material for wound healing. Contradictions in assessing different aspects of wounds.

them develop the ability to identify different wound states to support more effective treatment. Since the images are stored in the application, professionals can revisit them later for individual or collaborative reflection. In this way, they begin to change their working practices and gain ongoing opportunities for learning in their everyday care practice.

Excerpt 6 (Boundary Crossing Laboratory, session 5).

Speaker 5 (municipality nurse)	Is it macerated at the edges there? A little bit, I think, if you look northeast, or northwest, by the way, at the wound. Eh, there's a little spike sticking out.
Speaker 1 (nurse primary hcalth center)	Where?
Speaker 5	No, on the other side.
Speaker 1	There, yes there. Yes.
Speaker 5	There. Is it, it seems to be a little, now we're into the details, it's such a small difference.
Speaker 1	Yes.
Speaker 5	But is there a small macerated edge there?
Speaker 1	Yes, yes, that's it.
Speaker 5	Was it the same on the left side there too?
Speaker 1	Yes. So I, I thought about it a little bit, now afterwards. Because I've thought about this a little bit afterwards, when you're up in. That's what I also think is so interesting about this. It's because when you're up in the redeployment situation, you usually have to make pretty quick decisions.
Speaker 5	Mm, mm.
Speaker 1	But when you, because you have the pictures, you can think about it a little bit afterwards. And then, what I thought about afterwards, that, I think, exactly this. Yes, it's a little macerated there too.
Speaker 5	Mm.

Healthcare professionals have identified that the application is unable to assess all types of wounds. For example, it cannot accurately evaluate wounds where the visible cavity is smaller than the actual wound beneath the skin. In these cases, the app cannot measure the full extent of the wound. It also struggles to assess wounds where tendons are exposed. Moreover, wounds located in difficult areas – such as under a toe – often do not match reality when photographed, making the app unsuitable for their assessment. As a result, professionals request additional support or enhanced content that the app can provide. The application also offers limited guidance for wounds with exposed tendons, which require moist dressings. There is little information or assistance for these cases within the app.

This situation presents an opportunity for representatives from the app's development company to learn about these limitations and use this knowledge to enhance the application, ultimately improving wound care for a wider range of wound types.

Excerpt 7 (Boundary Crossing Laboratory, session 3).

Speaker 9 (company staff)	You talked about wounds that were not assessable with the app. I still think it's quite good to highlight that here.
Speaker 2 (municipality nurse)	Yes, I think we lifted it last time or made a wound that the patient takes where the entrance hole or whatever you should say, where the wound cavity is visible, but the wound is much larger. If you say so, there is a wound cavity under the entrance hole and then the app can't see or measure how big the wound is under the skin.
Speaker 5 (municipality nurse)	I also think there were concerns with the new patient who had a wound where a tendon was visible. We tried to use the app on that the other day and it was Lisa who was working on the app, but if I remember correctly, I couldn't get it into the app either.
Speaker 2	Was it that it couldn't take the picture, or?
Speaker 5	The picture just couldn't be uploaded. There was something wrong, I don't remember what it was, but there was a problem there too. It didn't work or it didn't look even in the app or it didn't match reality, if you put it that way.
Speaker 2	No, it's in a difficult place, the underside of the toe. So it's difficult to get it to work.
Speaker 4 (researcher)	With things that cannot be documented with the app, you can imagine that you would need extra support in some way or that there is someone with more knowledge about it that the app can help with.
Speaker 2	Yes, this is a wound where the tendon is exposed and it needs to be covered with a moist dressing. I don't think there was much information or help around that, I think.
Speaker 9	Do we have more perspectives on the use of the app and other difficulties that have arisen or something?
Speaker 5 (municipality nurse)	The app couldn't sense that it was a tendon, so it didn't see that it was a wound at all. Then I don't know what it was like. And then there's probably no such type of wound where tendons and bones are visible because it's a different wound dressing.
Speaker 9 (company staff)	Would it be possible to get a picture of that wound so that I could take this to the developers about where we are stuck a bit. That would have been good as a basis. Yes, to have this registered as a problem, so that we can troubleshoot.

The historicity, contradictions and expansive cycles in the interconnected activity systems are shown in Table 3.

How do the professionals learn?

Healthcare professionals find diagnosis to be challenging. The professionals participating in these collaborative sessions are nurses, not physicians, and nurses typically do not make formal diagnoses. However, in the context of wound care, nurses at primary care centres are often required to make indirect diagnostic judgements in order to treat and dress wounds appropriately. They must enter diagnostic codes into their systems as part of their daily responsibilities. Therefore, the professionals have expressed a need for the application to support them with appropriate coding for different wound diagnoses, to assist them in their routine tasks. Company representatives have now also recognised the importance of integrating diagnostic codes for various wound types into the application, to better support the professionals in their clinical workflows. Without this support, there is a risk that nurses will resort to using vague or generic codes.

The application offers support in several ways, both through the built-in knowledge base and through the structured assessment process that professionals are guided through when using the tool.

Table 3. Unit of analysis, historicity, contradictions and expansive cycles in the interconnected activity systems.

	Activity system as unit of analysis	Multi-voicedness	Historicity	Contradictions	Expansive cycles
Who are learning?	Interconnected activity systems Primary care center, municipality care, application developing company	Voices of primary care, municipality care, application developing company			
Why do they learn?			Health professionals have basic knowledge about wound healing. Professionals are facing problems in practice when wounds sometimes are hard to heal.	Contradictions between using the cheapest material for treatment in a short term and the most effective material for wound healing. Contradictions in assessing different aspects of wounds.	
What do they learn?	A new pattern of activity in assessing and treating wounds based on the use of the digital application.		Healthcare professionals ca discern image details and engage in reflection in ways that were not previously feasible. Certain categories of wounds remain particularly challenging to assess.	The application supports more structured observation and reflection in wound assessment. It also highlights the limitations of digital tools in evaluating certain types of wounds.	Healthcare professionals gain new opportunities to reflect on wound care, which contributes to improved treatment. Further development of the application could enhance wound care even more.

Excerpt number 8 (Boundary Crossing Laboratory, session 2).

Speaker 7 (municipality nurse)	It's good for me, who's not very good at wounds with different diagnoses. How to interpret them.
Speaker 1 (nurse primary hcalth center)	Mm, exactly. Yes, that's right.
Speaker 9 (company staff)	And you had input there last time that you actually wanted some of the most common ICD codes as well.
Speaker 1	I, from our side, it's really good support at the health center.
Speaker 9	Yes exactly. Mm, mm.
Speaker 1	Because that's how it is, we have, so it becomes, this thing with diagnoses is difficult. We're not used to, we who work with wounds are not used to making diagnoses. It doesn't come naturally to us to do that. And then you become very cautious when you don't feel sure about it. And then, then it becomes a kind of sloppy diagnosis, like skin wounds or leg wounds, or yes. Eh, but then I think that if you get a little guidance there, about the most common ones, of course, you can't be completely, maybe all, but the most common ones, I think that would be helpful, actually.
Speaker 6 (company staff)	Mm.

Using the application can sometimes be complicated in daily routines. One issue is the lack of consistent internet access in rural areas. Another challenge is that the municipality has not procured wound care materials from the company that developed the application, but instead from a different supplier. This creates difficulties for healthcare professionals when trying to adapt their practices, as materials from different companies have different product names.

It becomes hard to identify which products from the two suppliers correspond to each other in terms of properties. As a result, professionals need to spend time comparing materials across companies to determine equivalencies – those with the same characteristics. Using the application thus adds extra tasks, which the professionals find quite burdensome.

They would prefer that the procured materials come from the same company that supplies the application. This would greatly simplify their workflow when the app suggests a dressing material that is already available in the municipality's supply. Moreover, the professionals point out that various suppliers offer their own separate applications for wound care. Still, they wish that the app and the wound care materials would at least be provided by the same company.

Excerpt number 9 (Boundary Crossing Laboratory, session 6).

Speaker 6 (municipality nurse)	I've used it twice. Well, once it was really complicated. There was no coverage. And it just said chewed and chewed. And the second time, well, it went well then. But I, what I think has been very difficult is that we haven't procured material from Mölnlycke at the time. So it's been very difficult to keep thinking: -What kind of material do we have? So it's still been the same reshuffle that we usually do.
Speaker 4 (researcher)	Mm.
Speaker 6	So I think it would have been much smoother if we had had the same procurement. Because we, that is, the material we have, there has been almost nothing like that in the app.
Speaker 4	Have you been able to see for yourself that you have any similar products? That you can use? Have you made any comparisons with what you mentioned?
Speaker 6	Yes, we've looked at several and it's, I think, often the more Mepilex there is. Maybe it's just me who has thought that's what comes up when you put in. . . when you use the app.
Speaker 4	Mm.
Speaker 6	And yes, we only have two types of adhesive dressings like that. And then we have to think: —What should we put underneath that should be similar to what contains Mepilex?
Speaker 4	Mm.
Speaker 6	So I think it's been quite complicated, actually.
Speaker 4	Based on the product instructions in the knowledge library, you mean?
Speaker 6	Yes, so I think it's a bit complicated. I know, for example, because I was there and looked at Xerxes and those that we have procurement with.
Speaker 4	Mm.
Speaker 6	And it was much easier when you have the same material. Because then it became very easy to: -Yes, but then we'll take this dressing and that.
Speaker 4	Mm.
Speaker 6	Eh, then it was the same structure, I thought on the app that Xerxes had something similar to Ypsilon's. But from the outside, if we had had Ypsilon's procurement, it would probably have been easier.
Speaker 4	Mm. Do these Xerxes have any similar app that you use?
Speaker 6	No, it was because I took the wound training course this fall. And then it was at Xerxes there and lectured to us and told us about the app. So that's where I got to see it.
Speaker 4	Mm. But they have such an app too, then?
Speaker 6	Yes, they have.

Healthcare professionals generally perceive wound care as a complex and demanding aspect of their clinical work. In this context, the digital application is experienced as a valuable support, offering a structured workflow and practical suggestions that guide assessment and decision-making. The predefined steps within the app help users navigate wound evaluation systematically, reducing uncertainty and enhancing confidence in their clinical reasoning.

As professionals become more familiar with the application, they begin to recognise patterns across cases. When encountering patients with similar wound types, they can draw on prior experience and apply appropriate treatments more efficiently. The app also serves as a tool for confirmation, allowing users to validate their choices during the assessment process. This has led to improvements in the accuracy and speed of wound evaluations.

However, integrating the application into everyday routines has not been without difficulty. Time constraints, varying levels of digital competence, and the need to align new workflows with existing practices have presented practical challenges. Despite these, the professionals acknowledge that the app contributes positively to both individual learning and the overall quality of wound care.

Excerpt number 10 (Boundary Crossing Laboratory, session 7).

Speaker 4 (researcher)	Exactly. That's great. Uh, has this application affected your learning in any way, and if so, how?
Speaker 2 (municipality nurse)	Yes, to some extent. Considering, yes, that it is difficult with wounds and to get these tips or suggestions for dressings . . .
Speaker 4	Mm.
Speaker 2	. . . depending on what wound it is. So when you come to other patients who have similar wounds or what, how they have arisen and things like that. You already know that, yes, but then, this looks like the previous wound on that patient and then probably use these dressings. Will fit well or something.
Speaker 4	Mm.
Speaker 2	And then once you've run the app, you get a suggestion for what you've been thinking about or something.
Speaker 4	Mm.
Speaker 2	That way you can see if you've thought correctly or something.
Speaker 4	Exactly. Has it affected your assessment of the wound in any way?
Speaker 2	Yes.
Speaker 4	Mm.
Speaker 2	That's how it is.
Speaker 4	Yes exactly.
Speaker 2	So it has become better, or easier. It is difficult with wounds.
Speaker 4	Yes exactly.
Speaker 2	Many factors come into play, but. . . being able to form an opinion or an image more quickly then.
Speaker 4	Of the wound. Mm. Exactly. Uh, how has it been working using this application? Uh, have there been any challenges using it?
Speaker 2	Yes, it has been. From the very beginning, it was just a matter of getting into the routine. . .
Speaker 4	Mm.

Table 4 summarises the five principles of the interconnected activity systems.

Discussion

The professionals were involved in creating and implementing a radically new, broader object and concept for their activity, engaging in expansive learning by acquiring knowledge that did not yet exist (Engeström and Sannino 2010). Expansive learning offers a unique lens to analyse and promote learning for change (Engeström and Pyörälä 2021). Change Laboratory sessions demonstrated the collective capacity for transformative agency, enabling new possibilities for learning and change (Meijer et al. 2023). This led to a transformation in work practices and a shift in the culture of wound healing, facilitated by workshops addressing challenges in implementing a digital wound care app. This highlights the importance of including stakeholders in tailored, practice-integrated learning aligned with real healthcare challenges (Starke, Ludviga, and Fröhlich 2025).

This study explores how healthcare professionals' engagement with the digital wound care app altered their clinical perceptions and collaborative routines. Central to this were contradictions and double binds disrupting established practices and fostering reflection and innovation. To understand how these tensions led to learning, flow theory – focusing on timing, attentionality, and undergoing – was applied (Baygi, Introna, and Hultin 2021).

Table 4. Summary of the five principles in the interconnected activity systems.

	Activity system as unit of analysis	Multi-voicedness	Historicity	Contradictions	Expansive cycles
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How do they learn?		They learn through discussions with professionals from different healthcare organizations, with representatives from the company developing the app used to support wound care, and by using the app itself.		Learning is enhanced through app use and collaboration, improving reflection and wound care. Difficulties in assessing some wounds limit effective knowledge transfer and application, indicating a need for further app development.	By identifying contradictions in wound care practice, critical reflection and dialogue is prompted. Tensions drive collective efforts to reconsider and transform the approach, for qualitative improvements in assessment and treatment.

These concepts trace how knowledge and learning are dynamically enacted through lived, embodied, and intersubjective practice rather than discrete transfers.

Timing refers not just to scheduling but to the temporal alignment of actions within shared practice to seize emerging opportunities. In wound care, timing involves when and how professionals use the app to document and interpret wound data. Since wound healing is a temporal process, professionals develop sensitivity to its pace and care interventions. The app enables asynchronous collaboration-one documents, another interprets later – introducing new temporal demands. Misaligned timing, like late uploads or delayed feedback, disrupts collaboration and causes frustration. Learning happens as professionals renegotiate these temporal gaps and develop routines to synchronise actions. Through this, they enhance practical coordination and deepen their understanding of knowledge flow in a distributed system. Thus, timing is a key aspect of learning through temporal coordination.

Attentionality refers to how practitioners focus their perception and cognition to sense learning opportunities. The wound app reshapes what professionals attend to, visually and interpretively. Images over time reveal subtle wound changes often missed in daily practice. Professionals report 'seeing differently', noticing aspects like granulation or moisture in new ways. This perceptual shift is more than technical; it refines their clinical gaze. Attentionality is also co-constructed through discussions among professionals from different settings, aligning or contesting focus and creating shared interpretive repertoires. These subtle shifts redefine what is visible and relevant collectively. Thus, attentionality becomes a negotiated site of knowledge, showing clinical learning as relational and context-sensitive. In wound care, it helps professionals notice and respond to wound changes and patient needs, supporting reflective practice and informed decisions in real time.

Undergoing captures the experiential and emotional aspects of practice, reflecting how action transforms professionals and actualises learning opportunities. Engagement with the app exposed limitations in routines and assumptions, disrupting habitual interpretations and prompting clinical reconsideration alongside feelings of uncertainty and curiosity. Undergoing is active openness to change. Through repeated use and peer dialogue, professionals re-evaluated their wound care approaches and their role within a broader, interconnected care system. This transformation exemplifies expansive learning: not just acquiring knowledge, but fundamentally changing one's way of being.

Timing, attentionality, and undergoing reveal that learning emerges through practice itself. The wound care app is not merely an intervention but reshapes how knowledge is created and shared. Contradictions act as generative forces, driving professionals to rethink healthcare, from wound closure to shared interpretation, and its motives (Svensson 2020), from task completion to patient-centred understanding.

Double bind situations, where professionals feel constrained by existing routines yet need to act differently, become generative by pushing practitioners to reconceptualise the object of care. In wound care, this shift moves from a narrow clinical focus to broader shared sense-making, patient-centred documentation, and interprofessional learning. These changes emerge through iterative cycles of experimentation, feedback, and negotiation, aligned with the dynamics of flowing activity. This perspective shows how professional development, technological innovation, and organisational change are intertwined, highlighting the importance of collaboration in healthcare (Meijer

2023). Rather than viewing technology as a static tool, it is seen as evolving together with users and practices. The wound care app thus becomes a medium through which professionals reconfigure their practice, relationships, and understanding of meaningful care.

Conclusions

This paper has explored the learning processes triggered by the introduction of a digital application and how new forms of collaborative learning emerge across professional and organisational boundaries. Focusing on the implementation of a digital wound care tool, the study shows how healthcare professionals engage in expansive learning by collectively transforming their object of care and reorganising their activity system. These processes are driven by contradictions embedded in existing routines, which prompt reflection, negotiation, and innovation in everyday practice.

By drawing on expansive learning theory, the study contributes to research on professional learning by demonstrating how learning unfolds not as the adoption of predefined solutions, but as a bottom-up, practice-based process in which professionals actively redefine both their work and the technologies they use. In contrast to much of the digital transformation in healthcare, which remains largely top-down, the findings highlight how digital change can be shaped from within professional practice. This insight is particularly relevant in a healthcare context, but it may also be of interest to trade unions and professional associations concerned with how digitalisation affects professional autonomy, work organisation, and learning at work (cf. Lloyd and Payne 2025).

The study also advances the conceptual discussion within the theory on expansive learning by illustrating how expansive processes are enacted in increasingly digitalised work settings. The results suggest that core concepts such as object transformation, contradictions, and boundary crossing may need to be further elaborated to account for the temporal, attentional, and experiential dynamics of learning mediated by digital technologies. Here, the integration of flow theory adds analytical value by foregrounding how timing, attentionality, and undergoing shape learning as it unfolds in practice. The findings point towards a need to further develop expansive learning theory in dialogue with contemporary digital work environments.

At the same time, the study has limitations that should be acknowledged. The analysis is based on a single case and focuses on a specific type of digital tool within a particular healthcare context. Future research could explore similar bottom-up learning processes in other professional settings and across different types of digital technologies, as well as examine how such processes interact with more formal, top-down digitalisation initiatives. Longitudinal studies could also provide deeper insight into how expansive learning evolves over time as digital tools become stabilised or institutionalised.

This paper demonstrates that professional learning in digital transformation is best understood as an emergent, collective, and practice-based process. By foregrounding how healthcare professionals learn through engaging with digital technology in situ, the study contributes to broader debates on professional learning, digitalisation, and organisational change, and emphasises the importance of recognising bottom-up learning dynamics in the design and governance of digital transformation in healthcare.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributor

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